Diagnosing Human-to-Robot Transfer Failures with a Transfer Gap Budget in Simulation

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Abstract: A failed rollout does not show which stage of human-to-robot transfer failed. We introduce the *Transfer Gap Budget*, which replaces one pipeline component at a time with a declared oracle and measures the paired change in task success. A joint oracle measures non-additivity. In an injected-error study with 500 fresh seeds per arm, perception has the largest budget on both the UR5e and UR10e. A look-again correction, selected on development seeds, raises success by 26.8 and 25.8 pp, respectively, the largest observed gains among four corrections. In exploratory runs, perception stays the largest point-estimate budget when assumed errors are halved or doubled and when block size changes; when perception error shrinks, the budget picks a different correction that beats always applying look-again. On six demonstrations, diagnosis and a revised retreat raise paired success from 0–31/60 to 28–60/60 on unused seeds. Simulated depth perception meets pre-declared criteria on three manipulators. A G1 humanoid study finds 11/40 versus 31/40 placements for the same closed-loop controller across two runtime setups, with the cause unresolved. All robot evaluations are in simulation.

Keywords: human-to-robot transfer, failure diagnosis, oracle substitution

1 Introduction

Human-to-robot transfer requires motion estimation, retargeting, calibration, and executable timing. Aggregate success does not say which component to revisit, particularly with diverse human data [1]. We ask how much success correcting one pipeline component could recover.

The *Transfer Gap Budget* measures each component's net oracle gain. The gains can overlap. The largest development-set gain selects a correction for fresh evaluation. We test it on two arms, with real-video and depth-camera inputs, and report humanoid runtime sensitivity.

2 Related Work

Human-to-robot transfer. H2RBench reports sim–real agreement and quantitative transfer gains by execution stage on a Franka arm [2]. We intervene on pipeline components. Human-as-Humanoid and HumanEgo learn from human video [3, 4]; hand-pose quality and motion mismatch affect co-training [5, 6]. **Diagnosis and correction.** AHA, RoboFAC, and Guardian diagnose manipulation failures [7, 8, 9]. RHO and GaP search policy code and skill graphs [10, 11]. We use oracle gains to select from a fixed correction menu.

3 Method: the Transfer Gap Budget

Paired interventions. Let $\mathcal{K}$ contain perception, retargeting, tool-frame calibration, and execution timing. In the controlled-error study, *full* includes all four perturbations; $oracle_k$ removes only per-

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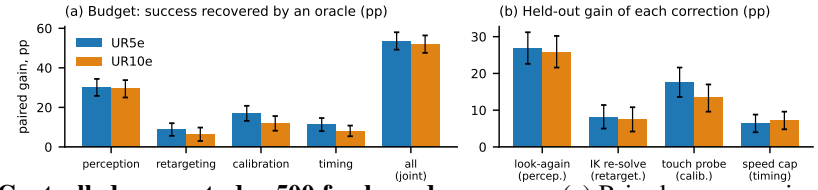


Figure 1: **Controlled-error study: 500 fresh seeds per arm.** (a) Paired success gains from single and joint oracles. (b) Gains from the four corrections; look-again was selected on development seeds. Error bars show paired bootstrap 95% intervals.

turbation k, and *oracle-all* removes all four. Their ideal outputs are exact object pose, unperturbed IK, zero tool offset, and nominal timing. Each random stream is keyed by seed and component, preserving initial scenes and unreplaced error draws across conditions. For binary success $s_c(i)$ on seed i in condition c,

$$B_k = \frac{1}{n}\sum_{i=1}^{n}\big(s_{\text{oracle}_k}(i) - s_{\text{full}}(i)\big), \qquad \Delta_{\text{int}} = \sum_{k\in\mathcal{K}} B_k - B_{\text{all}}. \tag{1}$$

Here B_{all} uses the joint oracle in the same paired difference. Budgets are signed net gains: an oracle can both rescue and lose trials. The residual shows whether summed single-stage gains exceed (> 0) or fall below (< 0) the joint gain; it does not allocate failures to interactions.

Correction selection. The menu pairs perception with look-again, retargeting with an IK re-solve, calibration with a modelled touch probe, and timing with a speed cap. We select $k^\star = \arg\max_k B_k$ on development seeds, breaking ties in pipeline order. Look-again and probing reduce assumed residual errors; their manoeuvres are not simulated.

Statistics. Paired percentile bootstrap intervals (95%; 5000 resamples in Section 5.1, 2000 elsewhere) resample seeds, preserving all condition scores of each seed. Demo intervals are descriptive: seeds 1000–1059 informed the revision, and demonstrations and stacks are not resampled. Gains use percentage points (pp).

4 Experimental Setup

Task. MuJoCo simulates a fixed-base UR5e or UR10e with a Robotiq 2F-85 from MuJoCo Menagerie [12], a 4 cm block with ± 3 cm position jitter, and a 14 cm tray. A scripted IK controller uses simulated contact in a UR5e-tuned layout. Success requires a lift of at least 5 cm, release within 3 cm of the goal, and speed below 0.02 m/s for 0.5 s after release.

A: controlled errors. The injected magnitudes are assumptions. Both arms receive the same seed-dependent draws: Gaussian xy perception error ($\sigma = 1.5$ cm), joint offsets ($\sigma = 0.5^\circ$ per joint), a uniformly directed tool-frame offset with magnitude $\mathcal{U}(0, 3)$ cm, and a timing speed-up of $\mathcal{U}(1.0, 4.5)$ relative to robot pace. The corrections reduce perception noise to 0.4 cm, joint-offset noise to 0.05°, or calibration noise to 2 mm per axis, or cap speed-up at $3\times$. Development seeds 0–59 select look-again on both arms. After freezing the protocol and four hypotheses, we evaluate ten conditions on 500 fresh seeds (3000–3499) per arm: *full*, four single oracles, the joint oracle, and four corrections applied to *full*.

E: sensitivity. On 200 further fresh seeds (4000–4199) per arm we repeat the ten conditions with injected errors scaled by 0.5 and 2, a 3 or 5 cm block, ± 5 cm jitter, and perception error cut to 25% alone and with calibration error doubled. Correction constants are unchanged; these runs are exploratory. To compare selection rules, the first 60 seeds choose the largest-budget stage and the other 140 evaluate its correction against always look-again and the best correction in hindsight.

B: real-video motion. Five pick-and-place cycles from two people in two freely licensed videos, plus one from a third person's 50 s recording, supply motion and timing. GEM-X [13] estimates hand trajectories; grasp and release frames are annotated manually. Reach and retreat are anchored to the simulated grasp and release points, and the carry path is rotated and scaled to connect them;

the scene is not reconstructed. Perception and calibration errors remain injected; no object detector is used, and retargeting is excluded from the real-video oracle analysis. Each demo–arm pair uses 60 seeds from the revision period (1000–1059) and 60 unused seeds (5000–5059). We report mean leg-wise path RMSE against the retargeted reference, after normalising time within each leg.

C: depth perception. A geometric estimator fits the table plane, clusters points above it, and selects a cube and tray using shape priors. Estimated position, yaw, and size drive the controller. Camera calibration is known nominally; stress tests inject errors. We test 5, 6, and 7 cm cubes on the two arms and a third manipulator (UR10e with a Robotiq 2F-140), with 40 seeds per size: 120 episodes per condition. **D: humanoid.** A Unitree G1 with Dex3 hands uses SONIC whole-body control [14]. We score grasp (5 cm lift with contact) and placement within 5 cm of the goal; we do not estimate stage budgets here.

5 Results

5.1 Controlled errors on two arms

Uncorrected success is 232/500 on the UR5e and 240/500 on the UR10e. Removing all errors recovers 53.6 and 52.0 pp. Perception has the largest budget: 30.0 [25.8, 34.4] pp on the UR5e and 29.4 [25.0, 33.8] pp on the UR10e. Calibration follows (17.0 and 12.0 pp), then timing (11.2 and 8.0 pp) and retargeting (8.8 and 6.4 pp); all single-stage intervals exclude zero (Fig. 1a).

The fresh sample gives positive retargeting budgets on both arms. The interaction residual is 13.4 [6.2, 20.8] pp on the UR5e and 3.8 [-2.6, 10.4] pp on the UR10e. The pilot's sign reversal is unsupported.

Look-again, selected from the development trials, raises success from 232 to 366/500 on the UR5e and from 240 to 369/500 on the UR10e. The paired gains are 26.8 [22.6, 31.2] pp and 25.8 [21.6, 30.2] pp, respectively. These are the largest observed gains. Probing gives 17.6 and 13.4 pp; IK re-solving and the speed cap 6–8 pp (Fig. 1b). Pairwise differences were not tested.

5.2 Sensitivity to assumptions and layout

Table 1: Sensitivity study, 200 seeds per arm (UR5e/UR10e); perception budget in pp.

Setting	Uncorrected	Perception	Largest
Nominal	100/101	+22.5/+29.0	perception
Errors $\times 0.5$	195/196	+2.5/+2.0	perception
Errors $\times 2$	13/11	+15.0/+16.5	perception
3 cm block	88/79	+23.0/+25.0	perception
5 cm block	83/82	+18.0/+22.0	perception
± 5 cm jitter	101/102	+22.5/+28.5	perception
Perception $\times 0.25$	149/157	-2.0/+1.0	timing/calibration
+ calibration $\times 2$	86/83	-4.0/+1.5	calibration

Perception has the largest budget in all twelve arm–setting cells under the nominal model, scaled errors and changed block size or jitter (Table 1). At $0.5\times$ errors uncorrected success is 195 and 196 of 200 and budgets are 2–2.5 pp. When perception error is cut to 25%, timing or calibration leads, and the rule's choice (timing on both arms) gains 8.6 and 10.7 pp on the held 140 seeds, against -1.4 and 0.7 pp for look-again; the best correction in hindsight gives 12.9 and 12.1 pp. With calibration error also doubled, the rule picks calibration and gains 44.3 and 47.1 pp, the best available, while look-again gives -6.4 and -1.4 pp. Where perception dominates, the rule coincides with always choosing look-again; the evidence is that its choice can change, not a general advantage.

5.3 Real-video demonstrations

The notebook demo fails uncorrected on both arms at about $6\times$ robot pace. Sequential diagnosis selects a $3\times$ speed cap, look-again, calibration probing, then a tighter $2\times$ cap. With the revised

Table 2: Successes of 60 paired seeds, uncorrected → revised stack; seeds 1000–1059 were reused in development, 5000–5059 were not. P1–P3 are demonstrators.

	Seeds 1000–1059		Unused seeds	
Demonstration	UR5e	UR10e	UR5e	UR10e
P1 notebook	0→ 57	0→ 60	0→ 60	0→ 60
P1 markers	0→ 51	0→ 25	0→ 49	0→ 28
P1 orange marker	0→ 46	0→ 58	0→ 51	0→ 59
P1 notepad	0→ 58	0→ 58	0→ 60	0→ 60
P2 box (own stack)	11→ 60	4→ 55	23→ 60	9→ 49
P2 box (P1 stack)	11→ 60	4→ 50	23→ 60	9→ 52
P3 recording (P1 stack)	35→ 59	30→ 56	31→ 57	21→ 53

retreat, the stack reaches 57/60 and 60/60 (Table 2). RMSE to the retargeted reference falls from 13.9 to 4.9 cm (UR5e) and 9.1 to 2.9 cm (UR10e), at the cost of slowing the motion.

The original stack scored 0/60 on all 3 additional P1 cycles on both arms, because their copied retreat paths intersect the tray after release; on P2 it reduced UR5e success from 11/60 to 0/60. We therefore added a 5 cm lift before retreating along the human path, selected on development seeds 0–19, at a path-RMSE cost of 2.7–4.1 cm. The revised P1 stack transfers to P2 and P3 without retuning; P1 markers remain difficult on the UR10e (25/60). Seeds 1000–1059 informed the revision, so those columns are exploratory. On seeds 5000–5059, which no development step used, every cell improves: the revised stack reaches 28–60/60 against 31 or fewer uncorrected, all paired intervals exclude zero, and P1 markers on the UR10e reaches 28/60.

5.4 Depth perception and humanoid runtime sensitivity

All 30/30 pre-declared acceptance cells reach the 90% point-estimate threshold (113–120 successes per 120 episodes). They cover yaw, distractors, depth noise, occlusion with a retry, and combined stressors on three manipulators. Report-only stress tests expose limits: a 2°/20 mm calibration error leaves the UR5e at 25/120, and arm occlusion without retry gives 0–16/120.

For the notebook path at robot pace, depth versus ground-truth success is 109 versus 108/120 (UR5e), 112 versus 111/120 (UR10e), and 120 versus 118/120 (third arm). Depth is 0.8–1.7 pp higher in these trials; path RMSE differs by less than 0.01 cm. All three meet the point-estimate margins of 10 pp success loss and 0.5 cm added RMSE (no inferential non-inferiority test). Combined depth stress gives 88/120, 88/120, and 91/120.

G1 humanoid. With IK-planned targets, the same frozen closed-loop controller places 11/40 objects in a volume-based installation and 31/40 in a packaged container (same GPU model, seeds 1200–1239; discordant placements 20 to 0, exact McNemar $p = 1.9 \times 10^{-6}$). Code and environment checks did not identify the cause; a four-seed MuJoCo 3.15.0 rebuild (0/4 lifts versus 4/4 on 3.14.0) also changed `filelock`.

6 Limitations and Conclusion

The study uses one simulated task, scripted controllers, a UR5e-tuned layout, and assumed error and correction magnitudes; the exploratory sensitivity runs do not vary the tray or table. Six demonstrations from three people and unreconstructed scenes limit generalisation; look-again and probing cost no simulated time. Where perception dominates, the budget rule cannot be told from always choosing look-again; only the reduced-perception settings separate them. The G1 discrepancy remains unexplained. Perception has the largest budget under the declared model and its correction the largest observed gain; reducing perception error moves both to other stages. This supports diagnosis in simulation; real-robot performance remains untested. An AI assistant helped draft parts of this manuscript and code; reported values were checked against saved results.

References

[1] Scaling H2R. CoRL 2026 Workshop "Scaling H2R" (Human-to-Robot). https://scaling-h2r-corl.github.io/, 2026. Call for papers, accessed Sept. 2026.

[2] C. Xiao, H. Zhan, S. Krishna, P. Meng, M. Z. Irshad, S. Zakharov, and D. Held. H2RBench: A real-to-sim benchmark for evaluating human-to-robot transfer. *arXiv preprint arXiv:2609.24778*, 2026. CoRL 2026.

[3] X. Lin, R. Yang, S. Lian, Z. Shen, B. Yu, C. Wu, H. Liu, Y. Zhang, H. Li, Q. Su, H. Liu, X. He, Y. Shi, C. Huang, Z. Zhang, B. Cheng, and K. Chen. Human-as-Humanoid: Enabling zero-shot humanoid learning from ego-exo human videos with human-aligned embodiments. *arXiv preprint arXiv:2606.32009*, 2026.

[4] Z. Wang, B. He, K. Yu, S. Lee, R. Gao, F. Huang, and Y. Aloimonos. HumanEgo: Zero-shot robot learning from minutes of human egocentric videos. *arXiv preprint arXiv:2605.24934*, 2026.

[5] R. Li, A. Prakash, A. Wen, S. Gupta, Y. Du, and P. Agrawal. What matters when cotraining robot manipulation policies on everyday human videos? *arXiv preprint arXiv:2606.06627*, 2026.

[6] Y. Domae, K. Shirai, H. Oh, R. Nakajo, T. Motoda, K. Makihara, M. Murooka, T. Yagi, Y. Bando, and R. Hanai. The embodiment gap in robot foundation models. *Transactions on Machine Learning Research*, 2026. arXiv:2608.18433.

[7] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo. AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation. *arXiv preprint arXiv:2410.00371*, 2024.

[8] Z. Ye, W. Lu, M. Ye, T. Lin, S. Yang, J. Yan, and B. Zhao. RoboFAC: A comprehensive framework for robotic failure analysis and correction. *arXiv preprint arXiv:2505.12224*, 2025.

[9] P. Pacaud, R. Garcia, S. Chen, and C. Schmid. Guardian: Detecting robotic planning and execution errors with vision-language models. *arXiv preprint arXiv:2512.01946*, 2025.

[10] K. Elmaaroufi, J. Svegliato, S. Kalade, G. Schelle, S. A. Seshia, and M. Zaharia. RHO: Your coding agent is secretly a roboticist. *arXiv preprint arXiv:2606.16458*, 2026.

[11] K. Chen, S. Xie, L. Fu, J. Yu, W. Pacini, S. Bajamahal, H. Kim, J. Drake, D. Kim, H. Xue, J. Francis, C. Juette, P. Schaldenbrand, M. Y. Seker, R. Wickramarachchi, U. Yoo, G. Wang, A. Murali, B. Sundaralingam, S. S. Sastry, S. Huang, Y. Zhu, L. Fan, and K. Goldberg. GaP: A graph-as-policy multi-agent self-learning harness for variational automation tasks. *arXiv preprint arXiv:2607.05369*, 2026.

[12] MuJoCo Menagerie. MuJoCo Menagerie. https://github.com/google-deepmind/mujoco_menagerie, 2026. Software repository.

[13] GEM-X. GEM-X: Monocular whole-body 3d human pose estimation. https://github.com/NVlabs/GEM-X, 2026. Software repository.

[14] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben, J. Park, D. Sami, Z. Wang, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. Fan, and Y. Zhu. SONIC: Supersizing motion tracking for natural humanoid whole-body control. *Science Robotics*, 11(117):eaed4592, 2026. arXiv:2511.07820.